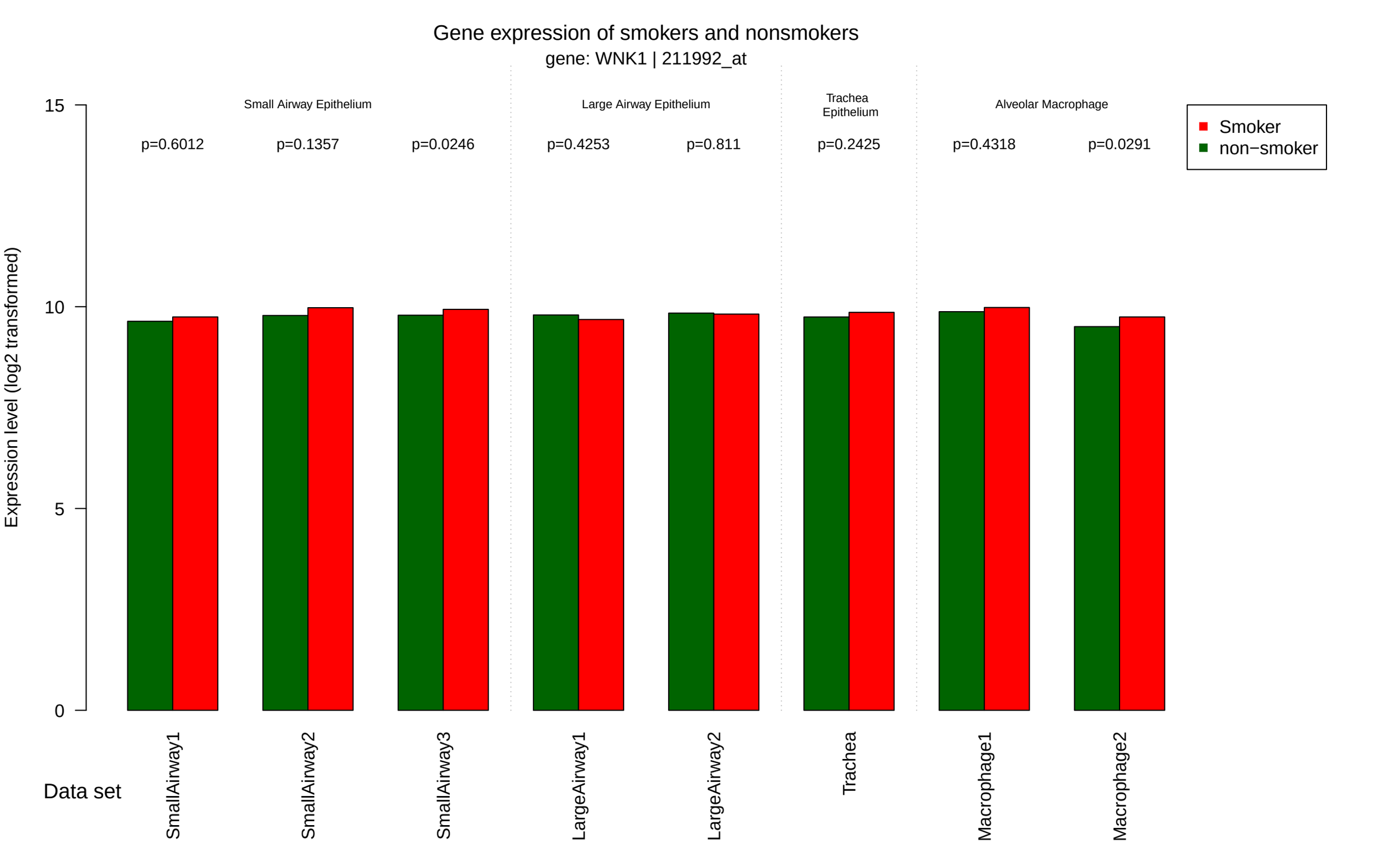
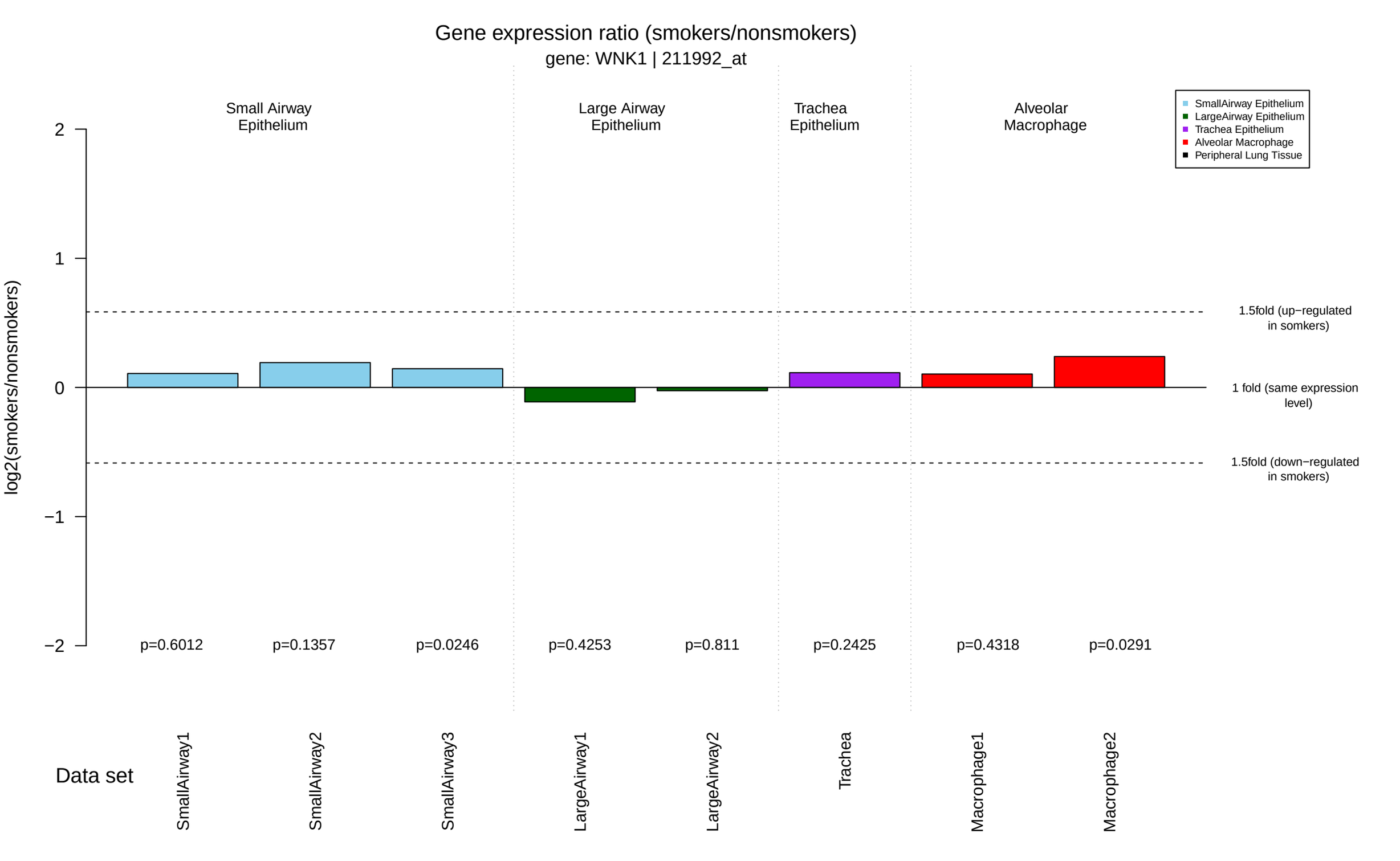
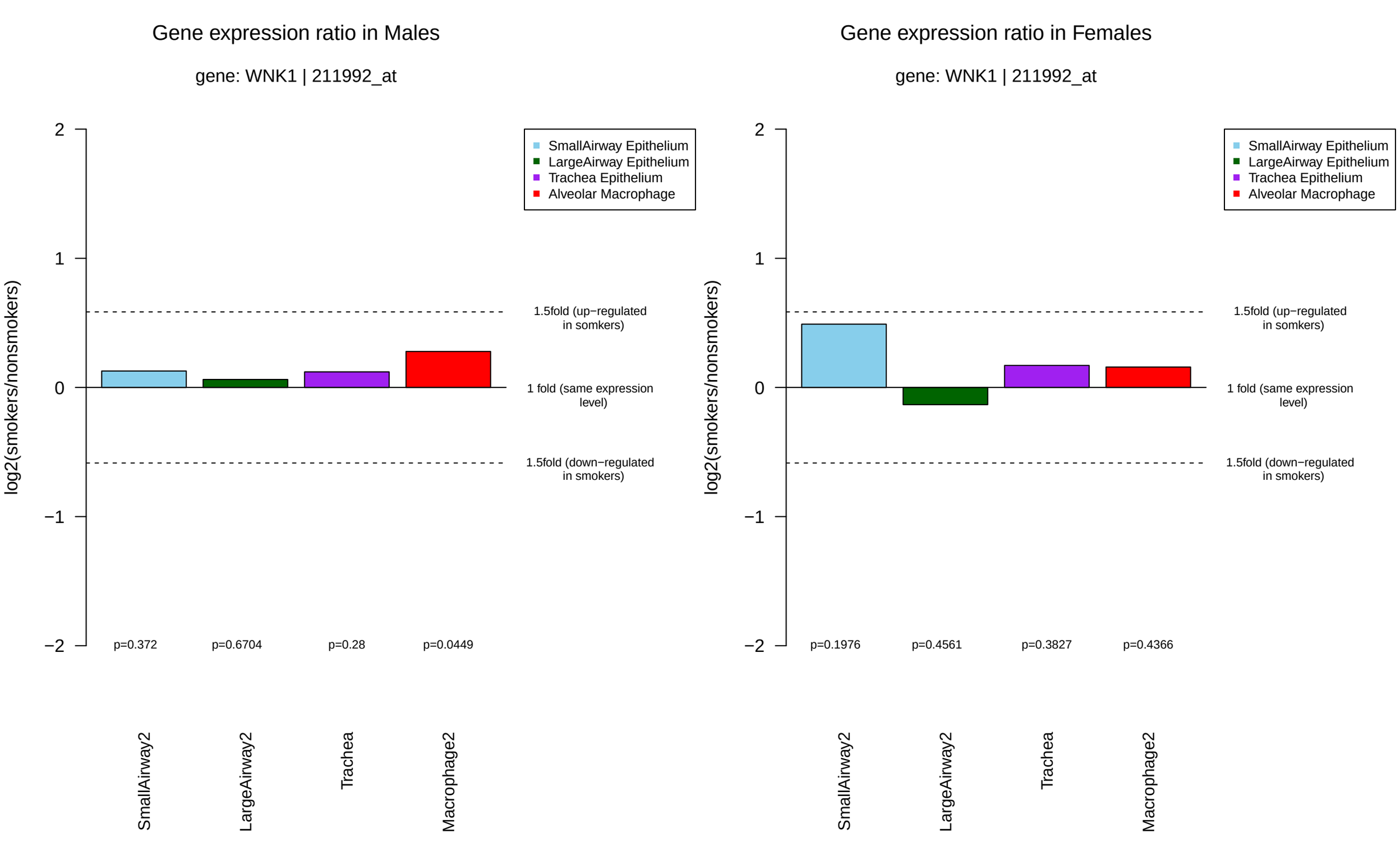
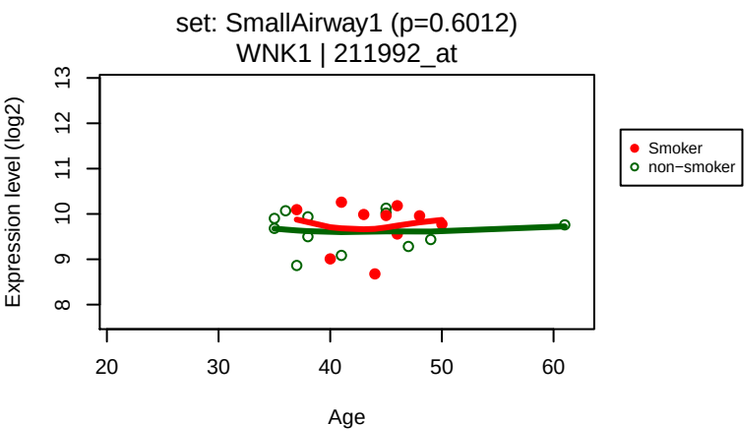
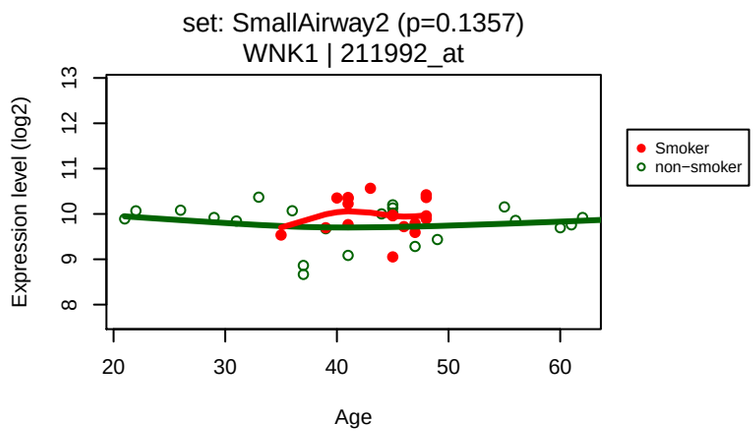
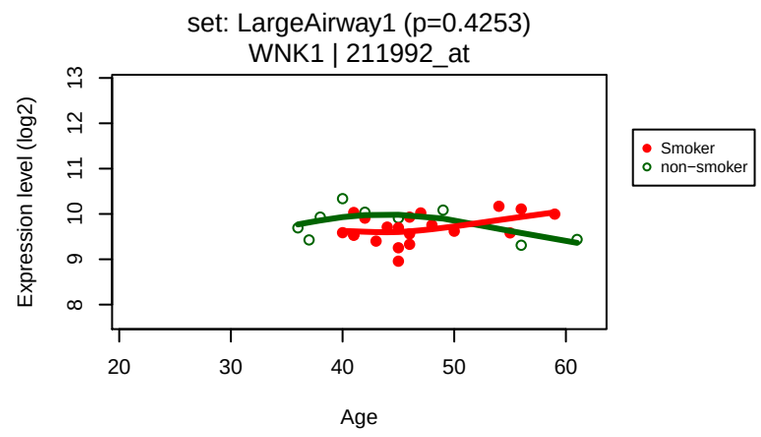

Gene expression of smokers and nonsmokers
gene: WNK1 | 211992_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6012
p=0.1357
p=0.0246
p=0.4253
p=0.811
p=0.2425
p=0.4318
p=0.0291
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

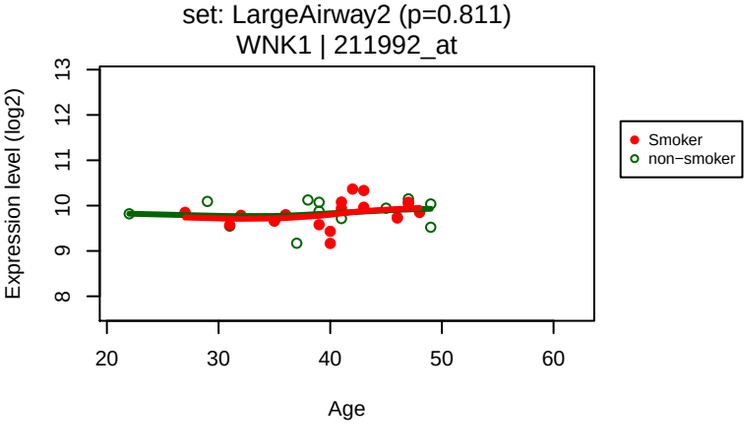

Gene expression ratio (smokers/nonsmokers)
gene: WNK1 | 211992_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6012
p=0.1357
p=0.0246
p=0.4253
p=0.811
p=0.2425
p=0.4318
p=0.0291
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

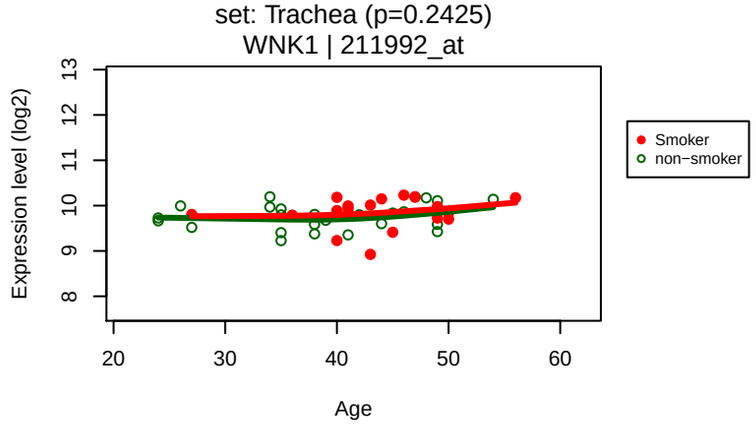

Gene expression ratio in Males
gene: WNK1 | 211992_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.372
p=0.6704
p=0.28
p=0.0449
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: WNK1 | 211992_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1976
p=0.4561
p=0.3827
p=0.4366
SmallAirway2
LargeAirway2
Trachea
Macrophage2

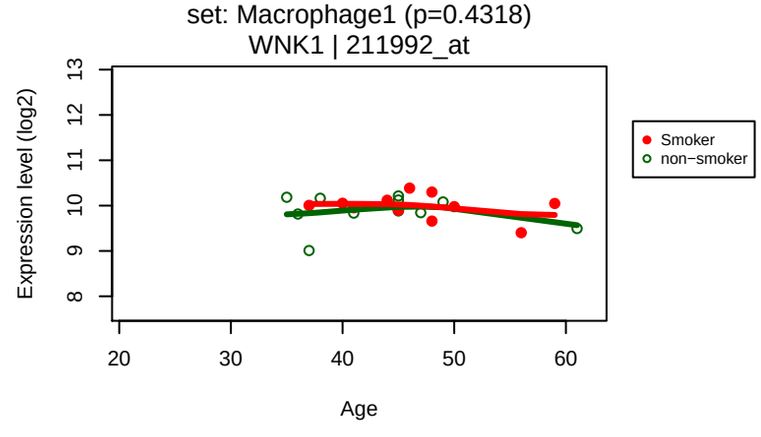

set: SmallAirway1 (p=0.6012)
WNK1 | 211992_at
Expression level (log2)
Age
Smoker
non−smoker

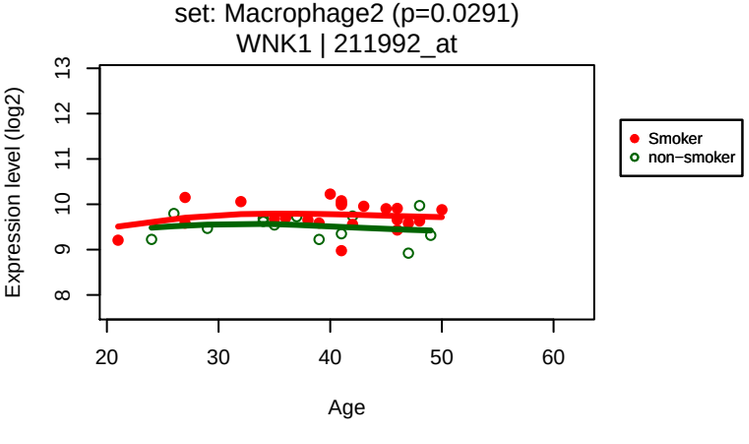

set: SmallAirway2 (p=0.1357)
WNK1 | 211992_at
Expression level (log2)
Age
Smoker
non−smoker

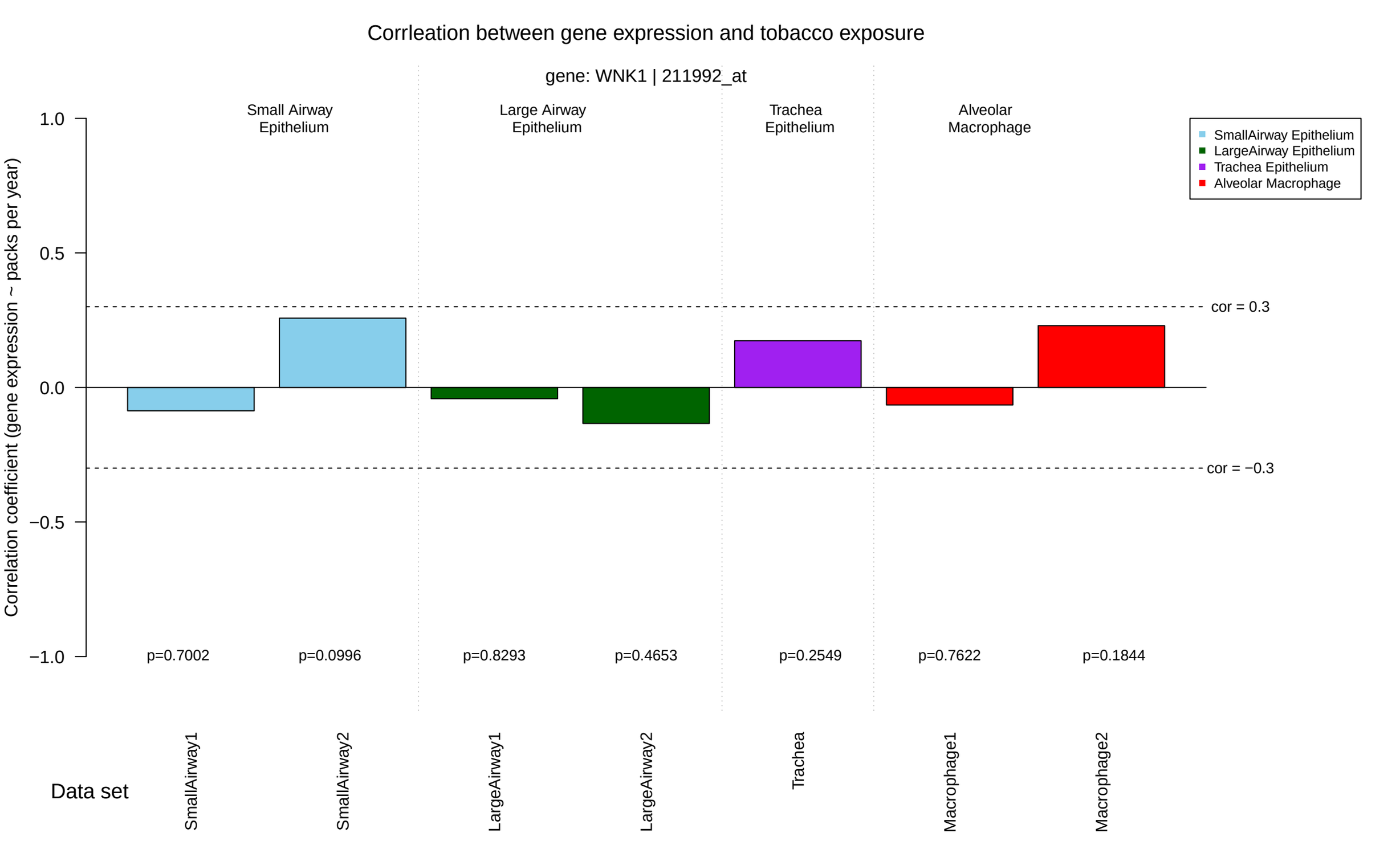

set: LargeAirway1 (p=0.4253)
WNK1 | 211992_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.811)
WNK1 | 211992_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2425)
WNK1 | 211992_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4318)
WNK1 | 211992_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0291)
WNK1 | 211992_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: WNK1 | 211992_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7002
p=0.0996
p=0.8293
p=0.4653
p=0.2549
p=0.7622
p=0.1844
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2